NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

Requests shall be referred to Headquarters, U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, MD 21005-5055, usarmy.APG.atec.mbx.atec-standards@mail.mil

Custodians:
Army – TE

Preparing activity:
ATC

AREA ENVR

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

REFERENCE FILE

Materiel Test Procedure 9-2-063
General Equipment Test Activity

2 August 1967

U. S. ARMY TEST AND EVALUATION COMMAND
COMMODITY ENGINEERING TEST PROCEDURE

CRANE TRUCK, WAREHOUSE

1. OBJECTIVE

The objective of this procedure is to determine the technical performance and safety characteristics of warehouse crane trucks and their associated tools and equipment, as described in Qualitative Materiel Requirements (QMR's), Small Development Requirements (SDR's), Technical Characteristics (TC's), and as indicated by the particular design and to determine the technical and maintenance suitability of the crane trucks for service tests.

2. BACKGROUND

New types of materials handling equipment are constantly being developed, and existing equipment is being improved or modified in order to support military requirements brought about by new tactical and logistical concepts of warfare. New technological design and materials and advanced concepts in handling, storing, loading, and unloading military supplies and equipment also contribute to the development of new equipment that can most efficiently and reliably perform assigned functions. Warehouse crane trucks are designed to perform certain functions that cannot be easily or reliably performed by other types of materials handling equipment.

3. REQUIRED EQUIPMENT

a. Tape for Measuring
b. Appropriate Pressure-type Indicating Gages
c. Tension Dynamometer
d. Stop Watch
e. Semicircular Templet (larger than half the circumference of the steering wheel)
f. Appropriate Loads
g. Scales for Weighing Loads, as Applicable
h. Motion Picture Camera with Film
i. Still Camera with Film
j. Lifting Equipment (must be capable of lifting the gross weight of the test item)
k. Four-part Lifting Sling (must be capable of lifting the gross weight of the test item)
l. Scribing Device (for scribing the turning radius of the test item on the ground)
m. Torque Arm (for attachment to a dynamometer)

4. REFERENCES

A. AMCR 385-1 Safety Responsibilities

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

B. AMCR 385-12 Verification of Safety of Materiel From Development Through Testing and Supply Disposition
C. AMCR 385-224 AMC Safety Manual
D. AR 70-10 Army Materiel Testing
E. AR 385-10 Army Safety Program
F. AR 705-5 Army Research and Development
G. MIL-STD-261 Test and Inspection of Crane Trucks, Warehouse, Gasoline, Pneumatic Rubber Tires
H. MIL-STD-262 Test and Inspection of Crane Truck, Warehouse, Electric, Solid Rubber Tires
I. MIL-C-3988 Crane Truck, Warehouse, Electric, Solid Rubber Tires
J. MIL-C-3998 Crane Trucks, Warehouse, Gasoline, Pneumatic Rubber Tires
K. MTP 2-2-500 Vehicle Characteristics
L. MTP 2-2-502 Inspection (Automotive)
M. MTP 2-2-503 Maintenance
N. MTP 2-2-803 Human Factors Engineering (Vehicle)
O. MTP 9-2-500 Physical Characteristics
P. MTP 9-2-501 Operator Training and Familiarization
Q. MTP 10-2-505 Human Factors Evaluation
R. TM 38-750 Army Equipment Record Procedures
S. TM 743-200 Storage and Materials Handling
T. USATECOM Reg. 385-6 Safety Release
U. USATECOM Reg. 700-1 Value Analysis in Materiel Testing
V. USATECOM Reg. 705-4 Equipment Performance Reports
W. USATECOM Reg. 705-12 Research and Development of Materiel

5. SCOPE

5.1 SUMMARY

This materiel test procedure describes the following tests conducted to evaluate warehouse crane trucks:

a. Initial Inspection - A study to determine the completeness of the test item, its physical characteristics and whether it meets safety requirements.

b. Pre-Operational Test - A study to determine whether performance characteristics of the test item comply with contractual requirements.

c. Operational Test - A study to determine whether or not the test item meets operational specifications.

d. Post-Operational Inspection - A study to determine the effects of testing upon the test item.

e. Maintenance Evaluation - A study to determine the maintenance requirements of the test item.

f. Safety Hazards Evaluation - A study to determine whether the test item has any safety hazards.

g. Human Factors Evaluation - A study to determine the ease of operating, accessibility of operating levers, etc., and adequacy of design as regards the operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

h. Value Analysis - A study to determine if the test item has any unnecessary features.

5.2 LIMITATIONS

This materiel test procedure provides general test procedures to be used as a guide to current testing technology for warehouse crane trucks. It does not include special testing techniques required for some special purpose crane trucks.

6. PROCEDURES

6.1 PREPARATION FOR TEST

6.1.1 Scheduling

6.1.1.1 Operator Training and Familiarization

Operators of the crane truck shall be trained in appropriate aspects of the crane truck as described in MTP 9-2-501 and the military standard for the test item.

6.1.2 Initial Inspection

Upon receipt, the test item shall be inspected as described in the applicable sections of MTP 2-2-502 (Automotive) and MTP 9-2-500 (Crane) and the following procedures:

6.1.2.1 Arrival Inspection

a. Record the following for the test item:

1) Nomenclature
2) Model No.
3) Serial No.
4) Manufacturer
5) Method of operation

b. Record the following for each major component, as applicable:

1) Nomenclature
2) Model No.
3) Serial No.
4) Manufacturer

c. Visually examine the test item and record and photograph the following, as required:

1) Materiel defects
2) Construction defects
3) Workmanship defects

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

d. Make and retain a copy of all manufacturers' labels and any operating or maintenance instructions attached to the test item.

6.1.2.2 Inventory Check

Conduct an inventory against the basic issue list item (BILI) and record any discrepancies as regards the maintenance package, spare parts, associated tools, associated equipment and components. Record all shortages.

6.1.2.3 Physical Characteristics

a. The vehicle portion of the test item shall be subject to the applicable sections of MTP 2-2-500.

b. The crane portion of the test item shall be subject to the applicable sections of MTP 9-2-500 and the determination of the following:

1) Method of operation (hydraulic, mechanical)
2) Type of test item (petroleum fueled, electric powered, towed)
3) Type of boom (telescopic, fixed, extension)
4) Maximum rated load
5) Type of outriggers (hydraulic adjustable, manually adjustable)
6) Type of lifting line
7) Dimension of lifting line
8) Boom topping system

6.1.3 Safety

Prior to conduct of testing, perform the following:

a. Ensure that all personnel involved in testing are familiar with safety procedures and requirements.

b. Ensure that the test item meets all safety standards.

c. When applicable, rectify features which do not comply with safety specifications. Record any action taken to make the test item meet safety standards.

d. When safety hazards exist and corrections can not be accomplished, halt the testing action.

e. Photograph all safety hazards with a still or motion picture camera, as applicable.

6.2 TEST CONDUCT

6.2.1 Pre-Operational Test

Prior to subjecting the test item to the operational test (paragraph 6.2.2) perform the following tests to determine whether performance characteristics of the test item comply with contractual requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

6.2.1.1 Clutch Pedal Test

a. Instrument the clutch pedal with a pressure-type indicating gage.
b. Determine and record the force required to depress the clutch pedal.

6.2.1.2 Steering Wheel Test

a. Position the test item on a level concrete surface with the steer wheels in the straight forward position.
b. Instrument the steering wheel with a tension dynamometer.
c. Determine and record the following:

1) The force required to turn the steering wheel
2) The number of turns of the steering wheel required to turn from one extreme direction to the other.

d. Block the steer wheels and apply a force having a value as described in the test plan to turn the steering wheel. Determine and record the following:

1) Ability of the steering mechanism to withstand the force applied
2) Force applied

6.2.1.3 Service Brake Pressure Sensitivity Towing Hook Test

NOTE: This procedure is conducted to determine the relationship between variable brake pressure and drawbar pull of the test item, expressed in percent of gross weight of the crane, with rated load, while the test item is being towed and to determine whether the towing hook has sufficient strength.

a. Prepare the test item as follows:

1) Install a pressure-type indicating gage on the brake pedal.
2) Attach a tension dynamometer to the towing hook of the test item.
3) Attach a towing cable from the tension dynamometer to a towing vehicle.
4) Load the test item with the rated load and record its gross weight.
5) Disengage the test item's engine and release the parking brake.

b. On a level concrete surface, with the loaded test item being towed, apply sufficient pressure to the brake pedal to produce a force recorded on the dynamometer approximately equal to 100 percent of the gross weight of the test item (with rated load) or as prescribed by the test plan. Repeat this procedure a minimum of 3 times.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

c. Record the following for each exercise:

1) Dynamometer reading
2) Applied brake pressure
3) Evidence of failure of the test item's towing hook

6.2.1.4 Service Brake Pressure Test

NOTE: This procedure is conducted to determine whether the brake pedal and brake system have the strength required by applicable specifications.

Perform the following:

a. Instrument the brake pedal with a pressure-type indicating gage.
b. On a level concrete surface, conduct a minimum of 3 stopping exercises by applying maximum pressure to the brake pedal.
c. Determine and record the following for each exercise:

1) Pressure applied to the brake pedal
2) Brake system failures

6.2.1.5 Load Line Hook, Boom Topping, and Sluing Tests

a. Perform the load line hook test as follows:

1) Set the boom at maximum operating elevation.
2) Attach the load line to the test item's rated load.
3) Hoist the load to maximum height and lower it back to the ground.
4) Record the following:

a) Height load was lifted
b) Time required to lift the load to the maximum height
c) Time required to lower the load from maximum height to the ground

NOTE: Time shall be monitored by a stop watch.

d) Any difficulties encountered

5) Repeat this exercise a minimum of 3 times.

b. Perform the boom topping test as follows:

1) Set boom at minimum operating elevation.
2) With the boom attached to the rated load, raise it to the maximum operating height and lower it back to the minimum operating elevation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

3) Record the following:

a) Boom minimum operating elevation
b) Boom maximum operating elevation
c) Time required to lift the boom to maximum elevation
d) Time required to lower the boom from maximum elevation to minimum operating elevation
e) Difficulties encountered

4) Repeat this exercise a minimum of 3 times.
5) Repeat steps 1) through 4) above, without the rated load.

c. Perform the sluing speed test as follows:

1) With the boom set at minimum operating elevation, attach the load line to the rated load and suspend it approximately 1 foot off of the ground.
2) Slue the boom through its applicable degrees of slue.
3) Record the following:

a) Test item's degree of slue
b) Time required to slue the boom through its applicable degrees of slue
c) Difficulties encountered

4) Repeat this exercise a minimum of 3 times.

6.2.1.6 Crane Speed Test

NOTE: This procedure is conducted to determine whether the test item can attain forward speed in high gear and reverse speed as required by the applicable specification.

a. Establish a speed course on a level concrete surface.

NOTE: Course shall have two reference points placed 88 feet apart from each other for measuring feet traveled per second.

b. Drive the test item (unloaded) at top forward speed, in high gear, over the test course between the reference points. Repeat this exercise using 3 different operators.

c. Record the following:

1) Direction of operation
2) Time required to pass between the reference points

d. Operate the test item (unloaded) in reverse, a minimum of three times, using different operators, and record the data of step c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

6.2.1.7 Acceleration Test

a. Establish an acceleration course on a level concrete surface .

NOTE: The course shall consist of a straight stretch of 44 feet from the starting to finishing points which shall be indicated by solid lines.

b. Place the front wheel of the test item on the starting line.
c. Accelerate an unloaded test item at its maximum rate, from standing start, over the test course and record the time required for the front wheels to cross the finish line.
d. Repeat steps b and c a minimum of 3 times using a different operator each time.
e. Repeat steps b through d using a test item carrying its rated load.

6.2.1.8 Acceleration Response Test

a. Position an unloaded test item on a level concrete surface with its engine idling.
b. Signal the operator to accelerate the test item, in a forward position, and record the time interval between depressing the accelerator pedal and the test item initial movement.
c. Repeat steps a and b a minimum of 3 times.
d. Repeat steps a through c with the test item in reverse.
e. Repeat steps a through d with a test item carrying its rated load.

6.2.1.9 Slope Ascension, Parking Brake, and Underclearance Tests

a. Drive an unloaded test item onto and up a ramp, stop half way up by use of the service brakes, and set the parking brakes.

NOTE:
1. Petroleum fueled test items shall negotiate a 15 percent ramp.
2. Electric (battery-powered) test items shall negotiate a 10 percent ramp.
3. Upon setting parking brakes, the service brakes are released.

b. Release the parking brakes and continue up the ramp until the next level floor is reached.
c. Observe the underclearance of the test item as it is driven from and onto the horizontal approaches at the top and bottom of the ramp.
d. Repeat steps a through c with the test item descending the ramp.
e. Repeat steps a through d in reverse.
f. Repeat steps a through e with the test item carrying its rated load.
g. Record the following for each exercise:

1) Direction of operation

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

2) Ramp ascension direction
3) Ability of parking brakes to hold on the ramp
4) Ability of test item to start from a complete stop on the ramp
5) Adequacy of the test item's underclearance at:

a) Top of ramp
b) Bottom of ramp

6) Difficulties encountered

6.2.1.10 Stopping Distance Test

a. Prepare for testing as follows:

1) Instrument the brake pedal with a pressure-type indicating gage.
2) On a level concrete surface, establish and mark a line where the operator will apply the brakes.

b. Drive an unloaded test item at maximum rated speed over the course and apply the brakes, when the line of step a.2 is reached, with maximum pressure on the pedal.

c. Record the following:

1) Pressure applied to brake pedal
2) Distance travelled by test item after application of the brakes to the stopping point.
3) Brake system failures

d. Repeat steps b and c using a minimum of 3 different operators.

e. Repeat steps b through d with the test item carrying its rated load.

6.2.1.11 Stability Test

a. Prepare for testing as follows:

1) Place the test item on a level surface
2) Attach the load line to a load equal to 110 percent of the maximum rated load.

NOTE: The load shall be compatible with the load chart for each 15° sluing position.

b. With the load placed in front of the test item, the boom at minimum operating elevation, lift the load until it clears the crane structures that could interfere with sluing of the load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

c. Slue the load 90 degrees to the right or left and hold in position for 1 minute.

d. Slue the load thereafter at 15 degree intervals and hold for 1 minute in each position throughout the entire slue range.

e. Repeat steps b through d with the boom being raised to mean and maximum boom operating elevation.

f. If the boom is telescopic, repeat steps b through f as follows:

1) Minimum operating length
2) 25 percent operating length
3) 50 percent operating length
4) 75 percent operating length
5) Maximum operating length

g. If applicable, repeat steps b through f using outriggers.

h. Record the following for each exercise:

1) Weight of load
2) Sluing angle
3) Boom operating elevation
4) As applicable, boom operating length(s)
5) Angles causing tipping
6) Evidence of possible instability and/or tipping
7) Failures

i. If failures occur at a specific boom position (elevation, length or slue), determine the allowable maximum load and repeat that portion of the test with the adjusted weight. Record the following:

1) Adjusted load weight
2) Boom operating elevation(s), length(s) or sluing position requiring adjusted load weights.

6.2.1.12 Lifting Attachments

a. Prepare the test item for testing as follows:

1) Fill all reservoirs, as applicable, i.e., fuel tank, hydraulic systems, cooling systems, etc., and record the gross weight of the test item.
2) Restrain test item from vertical movement on a level surface.
3) Attach a tension dynamometer to the end of equipment capable of lifting the gross weight of the test item.

b. Attach the tension dynamometer to one end of the test item's lifting eyes and apply a static force equal to the gross weight of the test item, hold it for one minute and record the following:

1) Evidence of deformation
2) Evidence of fractures
3) Permanent distortions

Source: https://assist.dla.mil -- Downloaded: 20260-26T14:54Z
Check the source to verify that this is the current version before use.

c. Repeat step b using each lifting eye.

d. Remove tension dynamometer and restraining device(s).

e. Attach a four-part lifting sling to the lifting equipment and eyes of the test item.

f. Lift the test item off of the ground and hold it, suspended for one minute, and record the following:

1) Evidence of deformation
2) Evidence of fractures
3) Permanent distortions

6.2.1.13 Wear of Steering Mechanism Evaluation

a. Prepare the test item for testing as follows:

1) Place a semicircular templet in front of the steering wheel.
2) Mark corresponding reference points on the upper edges of the steering wheel and the templet.

b. Turn the steering wheel counterclockwise until the steer wheels begin to turn. Mark the templet opposite the reference point on the steering wheel.

c. Repeat step b while turning the steering wheel clockwise.

d. Inspect the tie rod ends and other components of the steering linkage for excessive wear.

e. Record the following:

1) The distance between the two marks on the templet
2) Evidence of excessive wear to tie rod ends and other steering linkage components.

6.2.1.14 Structural Load and Overload Tests

a. Prepare the test item as follows:

1) Block the test item off of the ground on a level surface.
2) Counterweight or anchor the test item to prevent tipping.
3) Install pressure gages to the crane's hydraulic system.

b. Lift a load equal to 200 percent of the rated load and slue it clockwise 15°, 30° and 45° and then counterclockwise 15°, 30° and 45° and hold in each position for 5 minutes. Inspect and record any evidence of damage.

c. Lift a rated load with the test item and pull it 3 feet outward from the bumper and allow it to swing abruptly against the bumper, and then slue it clockwise 15°, 30° and 45° and then counterclockwise 15°, 30° and 45°. Determine and record the following:

1) Damage sustained when swinging the load against the bumper
2) The maximum pressure required to manipulate a rated load through sluing ranges

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

3) The applied percentage of maximum pressure required to manipulate the load for each function

d. For a style III crane truck, drive the test item a minimum of 400 feet, with a load equal to a minimum of 110 percent of the rated load, positioned on the deck of the truck. Inspect the test item and record evidence of damage.

e. Using the load hook, lift a load equal to 300 percent of the rated load at the closest load rating position on the longitudinal centerline of the crane and hold it for one minute. Inspect the test item and record evidence of damage.

NOTE: Support the boom for this test.

6.2.1.15 Turning Radius and Travel Capacity Tests

a. Instrument the test item with a device capable of scribing a line on a level surface at the vehicle's outermost position of swing.

b. Drive the test item, with its steer wheels cramped as far as possible clockwise, through a 360 degree turn over a level surface. Measure and record the scribed circle.

c. Repeat step b going counterclockwise.

d. Freely suspend a rated load from the hook at the test item and drive it at low speed for a minimum of 100 feet. Record any difficulties encountered.

6.2.1.16 Power Train Static Torque Test

NOTE: The power train static torque test is conducted to determine whether the drive line of the crane truck can withstand a specified percent of the maximum bare engine torque or stalled engine torque.

a. Clamp a torque arm (which is attached to a dynamometer) to the input shaft of the test item's transmission or to the output shaft of the torque connecter, as applicable.

b. With the transmission in the lowest gear range, lock the drive wheels and gradually apply force until 110 percent of the rated bare engine or engine stall torque is reached. Disassemble the transmission and examine it for failures, fractures and permanent distortion and deformation.

c. Record the following:

1) Force applied to the torque arm
2) Evidence of transmission failures, fractures and permanent distortion and deformations.

6.2.1.17 Controls Test

NOTE: The controls are tested to determine whether they meet the requirements of the applicable specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

a. For power-shifted transmissions only, place the direction control lever in each drive position and attempt to start the engine with the control in each position and record the ability of the engine to start in all of the drive positions.

b. For all transmissions, place the control lever in neutral, start the engine, and record the ability of the engine to start.

c. The boom topping, boom extension, boom slue and load line controls shall be operated individually and simultaneously for appropriate operations. Record the capability of the test item to perform appropriate operations while using the boom topping, boom extension, boom slue and load line controls, individually and simultaneously.

6.2.1.18 Hook and Cable Tests

NOTE: The hook and cable tests are conducted to determine whether hook clearances, cable reach, and paying out of the hook comply with the applicable specification.

a. Place the test item on a level surface 5 feet above the ground.

b. Fully extend and top the boom (its maximum operating length and height) with the hook in full raised position.

c. Lower the hook to the surface on which the test item is positioned and then lower it another 5 feet. Determine and record the following:

1) Distance of clearance between suspended cable and test item
2) Distance from the top of the boom to the ground
3) Ability of the cable to extend 5 feet below where the test item was positioned.

d. With the hook in full raised position, place the boom in mean and then minimum operating positions and record the distance from the ground to the pull point of the hook at each position.

6.2.2 Operational Test

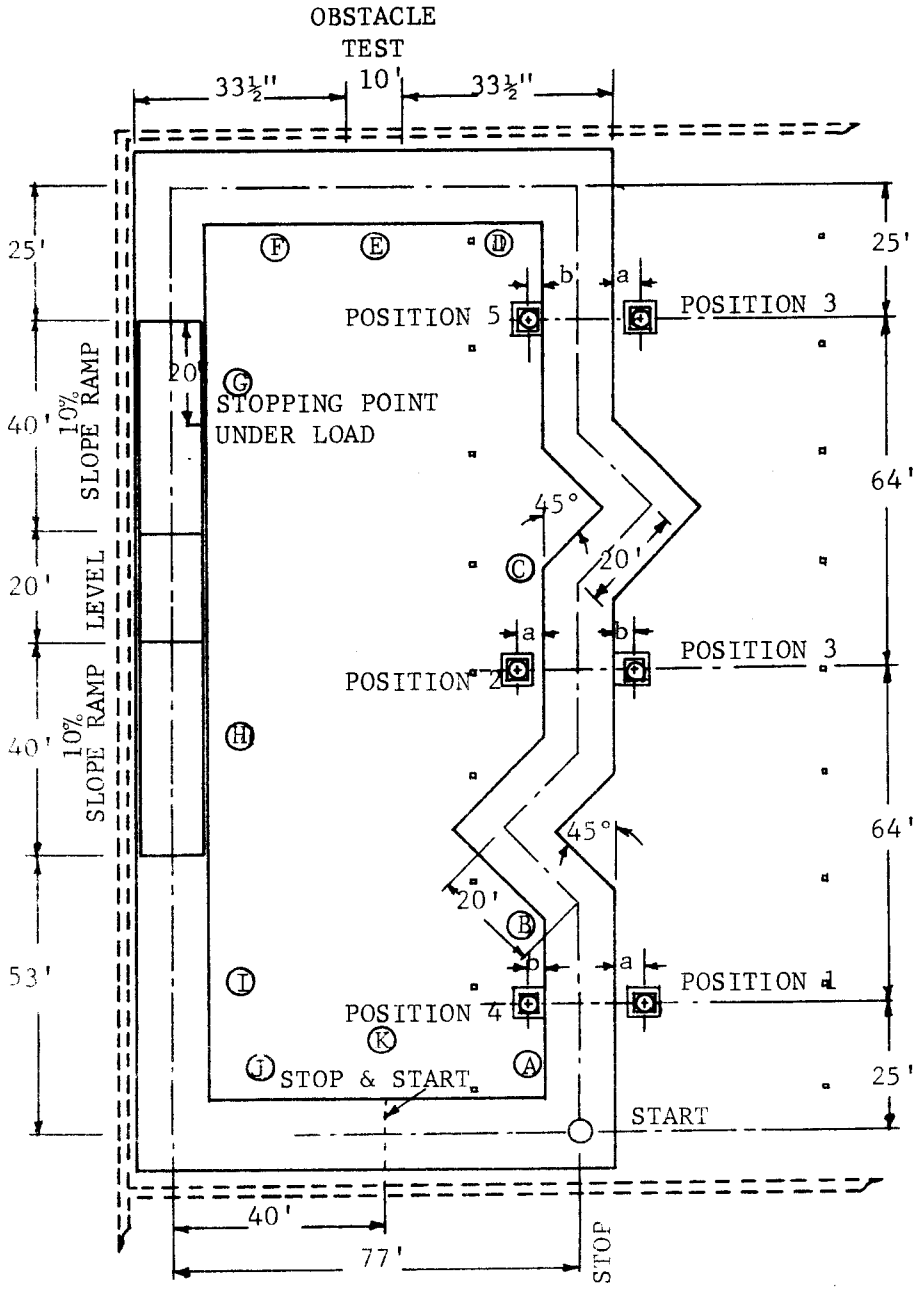
The operational test shall be conducted as described in applicable sections of MIL-STD 261 and MIL-STD 262 or other appropriate military standard.

NOTE: 1. Figure 1, supported by figures 2 and 3, describes the special test course as shown in MIL-STD 261 and MIL-STD 262.
2. The test item will perform a series of operations on a continuous basis during each 8-hour operating day to provide accelerated use of the test item.
3. The test item will negotiate 1200 complete circuits of the test course, half of which shall be in a clockwise direction and half in a counterclockwise direction.

6.2.3 Post-Operational Inspection

At the completion of operational testing, the test item shall be subjected to technical and pre-operational inspections as described in

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

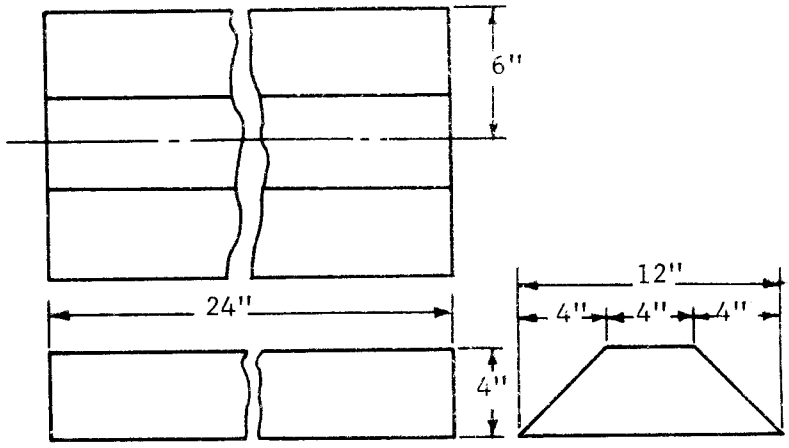


NOTES:
a = 60" FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD
b = 36" FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD
LIGHTING TO BE A MINIMUM OF 5 FOOT CANDLES AT OPERATING LEVEL
TEST COURSE IS FINE-FLOAT FINISH CONCRETE OR EQUIVALENT
TEST COURSE IS SHOWN LAID OUT LENGTHWISE IN A STANDARD 200' BY 600' NAVY WAREHOUSE
Ⓐ, ETC. = SECTION DESIGNATION

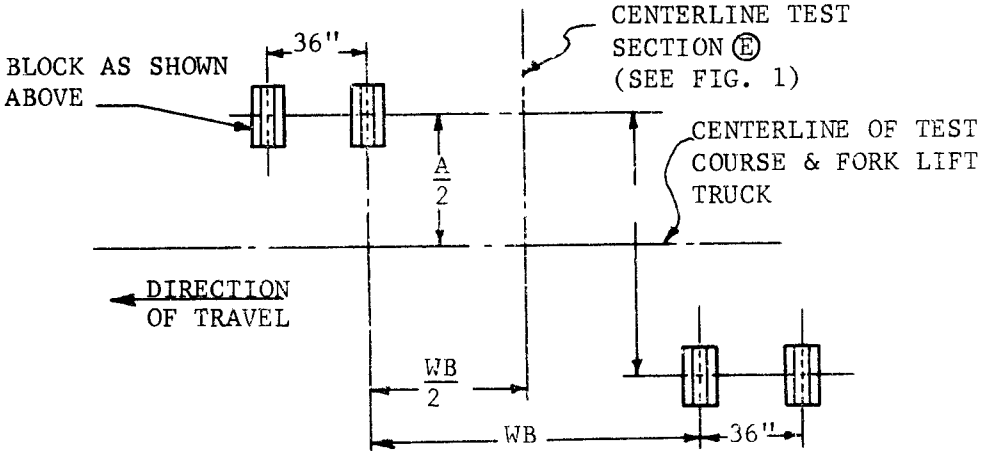
Figure 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

OBSTACLE BLOCK CONSTRUCTION



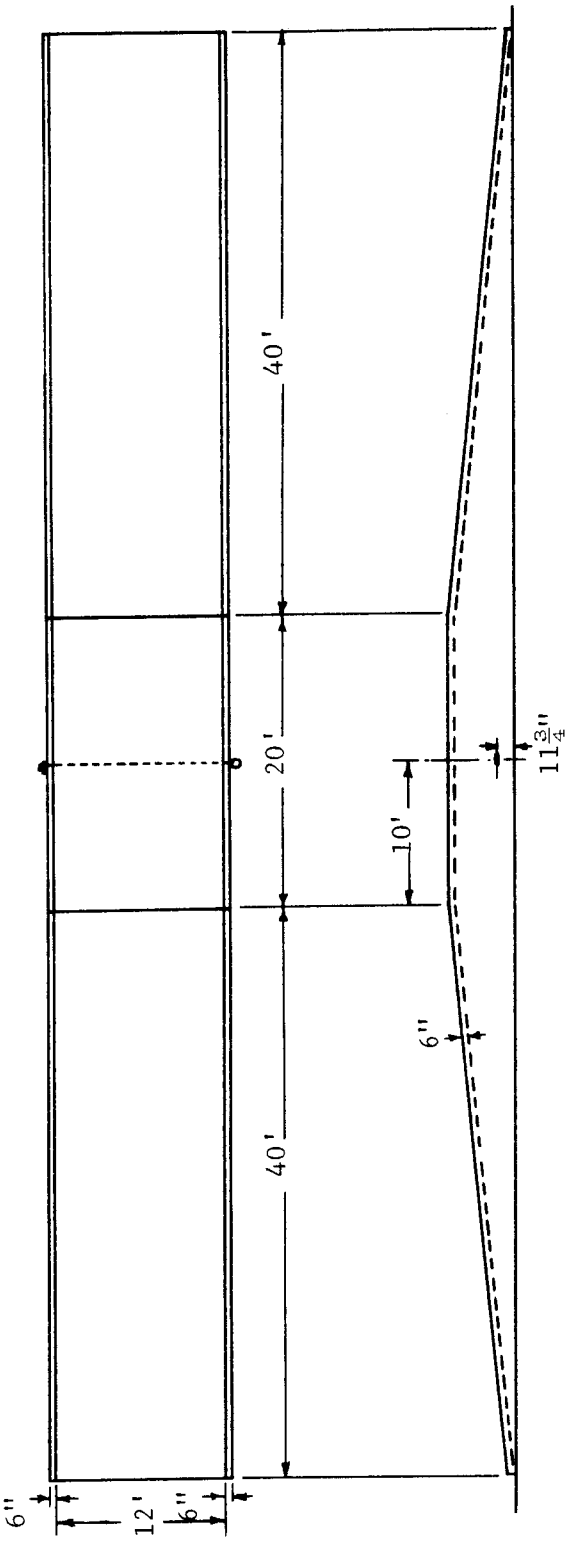
LAYOUT FOR OBSTACLE COURSE



NOTES:
1. "A" TREAD OF DRIVING WHEELS
2. "WB" WHEEL BASE OF TRUCK UNDER TEST

FIGURE 2 - DETAIL OF OBSTACLE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.



SPECIFICATIONS:
RAMP TO BE 6 INCH CONCRETE PAD, FINE FLOAT FINISH ON COMPACTED FILL
CURBING TO BE REINFORCED
EYE TO BE SUNK IN WALL OF RAMP TO TAKE A 7500 POUND DYNAMIC DRAWBAR PULL LOAD

DRAWING "G" DESIGN

Figure 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

MTP 2-2-502 (automotive) and MTP 9-2-500 (crane) to determine adverse effects caused during the test conduct and a measurement of the steering mechanism as described in paragraph 6.2.1.13.

6.2.4 Maintenance Evaluation

a. During the conduct of testing, the maintainability of the automotive portion of the test item shall be determined as described in MTP 2-2-503.

b. Throughout the test a log will be maintained, for the crane portion of the test item, in which the following observations and maintenance operations will be recorded:

1) Scheduled maintenance conducted in accordance with manufacturer's instructions furnished with the test item
2) Equipment deficiencies, causes, and suggested or corrective action taken
3) The adequacy of spare parts and tools
4) The adequacy and accuracy of the technical and maintenance instructions provided by the manufacturer

6.2.5 Safety Hazards Evaluation

Determine the safety characteristics of the test item, based on the safety statement issued by the developing agency, as follows:

a. At all times during the conduct of the test, normal safety precautions for material handling equipment and warehouse operations will be exercised.

b. Throughout the test period, test personnel will observe and record any condition or procedure that might present a safety hazard, the cause of the hazard, and steps taken to alleviate the hazard(s).

c. Upon completion of testing, a safety release recommendation shall be developed as prescribed in USATECOM Regulation 385-6.

6.2.6 Human Factors Evaluation

The effectiveness of the man-item relationship during use of the test item and the degree of ease, simplicity, and effort in operating, servicing, and maintaining the test item shall be evaluated as described in applicable sections of MTP 2-2-803 and MTP 10-2-505.

6.2.7 Value Analysis

Determine whether or not the test item has any nonfunctional, costly, or nice-to-have features as described in USATECOM Regulation 700-1.

a. During operation and maintenance of the test item, observations shall be made for features which could be eliminated without adversely affecting essential performance requirements, reliability, durability, or safety.

b. Question test personnel and test director for features of the test item which may be eliminated without decreasing the function value of the test item.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

NOTE: Comments will be separately entered in the daily log.

c. Record the following:

1) Nonfunctional features
2) Costly features
3) Nice-to-have features
4) Comments of test personnel and test director

6.3 TEST DATA

6.3.1 Preparation for Test

6.3.1.1 Initial Inspection

Data shall be collected and recorded as described in applicable sections of MTP 2-2-502 (automotive) and MTP 9-2-500 (crane), and the following:

6.3.1.1.1 Arrival Inspection -

a. Record the following:

1) For the test item:

a) Nomenclature
b) Model No.
c) Serial No.
d) Manufacturer
e) Method of operation

2) For each major component: (as applicable)

a) Nomenclature
b) Model No.
c) Serial No.
d) Manufacturer

3) Results of visual examination:

a) Material defects
b) Constructional defects
c) Workmanship defects

b. Retain all photographs, and copies of manufacturer's labels and operating or maintenance instructions.

6.3.1.1.2 Inventory Check -

Record all shortages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

6.3.1.1.3 Physical Characteristics -

a. For the automotive portion of the test item, data shall be collected and recorded as described in the applicable sections of MTP 2-2-500.
b. For the crane portion of the test item, data shall be collected and recorded as described in the applicable sections of MTP 9-2-500, and the following:

1) Method of operation (hydraulic, mechanical)
2) Type of test item (petroleum fueled, electric powered, towed)
3) Type of boom (telescopic, fixed, extension)
4) Maximum rated load
5) Type of outriggers (hydraulic adjustable, manually adjustable)
6) Type of lifting line
7) Dimension of lifting line
8) Boom topping system (hydraulic, cable)

6.3.1.2 Safety

a. Record the following:

1) Noted safety hazards
2) Action taken to rectify safety hazards

b. Retain all photographs

6.3.2 Test Conduct

6.3.2.1 Pre-Operational Test

6.3.2.1.1 Clutch Pedal Test -

Record the force, in psi, required to depress the clutch pedal.

6.3.2.1.2 Steering Wheel Test -

Record the following:

a. The force, in psi, required to turn the steering wheel
b. The number of turns of the steering wheel required to turn from one extreme direction to the other.
c. Ability of the steering mechanism to withstand the force applied
d. Force, in psi, applied when steer wheels were blocked

6.3.2.1.3 Service Brake Pressure Sensitivity Towing Hook Test -

a. Record the gross weight of the test item with rated load, in pounds.
b. Record the following for each exercise:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

1) Exercise (1, 2, 3)
2) Dynamometer reading, in psi
3) Applied brake pressure, in psi
4) Towing hook failure

6.3.2.1.4 Service Brake Pressure Test -

Record the following for each exercise:

a. Exercise (1, 2, 3)
b. Maximum pressure applied to the brake pedal, in psi
c. Brake system failures

6.3.2.1.5 Load Line Hook, Boom Topping, and Sluing Tests -

Record the following for each exercise:

a. For load line hook test:

1) Exercise (1, 2, 3)
2) Height load was lifted, in feet
3) Time, in seconds, required to lift the load to maximum height
4) Time, in seconds, required to lower load from maximum height to the ground
5) Any difficulties encountered

b. For boom topping test:

1) Exercise (1, 2, 3)
2) With, without rated load
3) Boom minimum operating elevation, in degrees
4) Boom maximum operating elevation, in degrees
5) Time, in seconds, required to lift the load to maximum elevation by means of raising the boom
6) Time, in seconds, required to lower the load from maximum elevation to minimum operating elevation by means of lowering the boom
7) Difficulties encountered

c. For sluing speed test:

1) Exercise (1, 2, 3)
2) Test item's degrees of slue
3) Time, in seconds, required to slue the boom through its applicable degrees of slue
4) Difficulties encountered

6.3.2.1.6 Crane Truck Speed Test -

Record the following for each exercise:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

a. Operator (1, 2, 3)
b. Direction of operation (forward, reverse)
c. Time, in f.p.s., required to pass between the reference points

6.3.2.1.7 Acceleration Test -

a. Operator (1, 2, 3)
b. Loaded, unloaded
c. Time, in seconds, required to travel from the starting line to the finish line

6.3.2.1.8 Accelerator Response Test -

Record the following for each exercise:

a. Exercise (1, 2, 3)
b. Loaded, unloaded
c. Direction of operation (forward, reverse)
d. Time interval, in seconds, between the acceleration signal and the instant the test item started to move.

6.3.2.1.9 Slope Ascension, Parking Brake, and Underclearance Tests -

Record the following for each exercise:

a. Loaded, unloaded
b. Direction of operation (reverse, forward)
c. Ramp ascension direction (up, down)
d. Ability of parking brakes to hold on the ramp
e. Ability of test item to start from a complete stop on the ramp
f. Adequacy of the test item's underclearance at:

1) Top of ramp
2) Bottom of ramp

g. Difficulties encountered

6.3.2.1.10 Stopping Distance Test -

Record the following for each exercise:

a. Operator (1, 2, 3)
b. Loaded, unloaded
c. Maximum pressure, in psi, applied to brake pedal
d. Distance travelled, in feet, by test item after application of the brakes to the stopping point
e. Brake system failures

6.3.2.1.11 Stability Test -

Record the following for each exercise:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

a. Outriggers (used, not used)
b. Weight of load, in pounds
c. Sluing angles, in degrees
d. Boom operating elevation, in degrees
e. As applicable, boom operating length(s), in feet
f. Angles causing tipping
g. Evidence of possible instability and/or tipping
h. Failures
i. For repeated operations due to failures:

1) Adjusted load weight, in pounds
2) Boom operating elevation(s), in degrees
3) Boom operating length(s), in feet
4) Sluing position, in degrees

6.3.2.1.12 Lifting Attachments -

Record the following:

a. Gross weight of the test item with rated load, in pounds
b. Tension results for each lifting eye:

1) Evidence of deformation
2) Evidence of fractures
3) Permanent distortions

c. Results of test item being completely suspended:

1) Evidence of deformation
2) Evidence of fractures
3) Permanent distortions

6.3.2.1.13 Wear of Steering Mechanism Evaluation -

Record the following:

a. Distance between the two marks on the templet, in inches
b. Evidence of excessive wear to tie rod ends and other steering linkage components

6.3.2.1.14 Structural Load and Overload Tests -

Record the following:

a. The maximum pressure, in psi, required to manipulate a rated load through each sluing range
b. The applied percentage of maximum pressure required to manipulate the load for each function
c. Damages sustained when:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

1) Sluing 200 percent of the rated load
2) Swinging the load against the bumper
3) When applicable, carrying a load 400 feet on the test item's deck
4) Handling 300 percent of the rated load on the longitudinal centerline of the crane

6.3.2.1.15 Turning Radius and Traveling Capacity Tests -

Record the following:

a. Radius, in feet, of the turning circle with the test item going in a:

1) Clockwise direction
2) Counterclockwise direction

b. Difficulties encountered when carrying a rated load, suspended from the test item's hook, for 100 feet

6.3.2.1.16 Power Train Static Torque Test -

Record the following:

a. Force applied to shaft in ft-pounds
b. Evidence of transmission failures, fractures and permanent distortion, and deformations

6.3.2.1.17 Controls Test -

Record the following:

a. For power-shifted transmissions:

1) Ability of the engine to start with the control lever placed in each of the drive positions

b. For all transmissions:

1) Ability of engine to start with control lever in neutral

c. Capability of the test item to perform appropriate operations while using the boom topping, boom extension, boom slue and load line controls, individually and simultaneously.

6.3.2.1.18 Hook and Cable Tests -

Record the following:

a. Distance of clearance, in feet, between suspended cable and test item

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

b. Distance, in feet, from the top of the boom to the ground.
c. Ability of the cable to extend 5 feet below where the test item was positioned
d. Distance, in feet, from the ground to the pull point of the hook (in full raised position) with the boom in:

1) Mean operating position
2) Minimum operating position
3) Maximum operating position

6.3.2.2 Operational Test

Data shall be collected and recorded as prescribed in the applicable section of MIL-STD 261 and MIL-STD 262 or other appropriate military standard.

6.3.2.3 Post-Operational Inspection

a. Data shall be collected and recorded as prescribed in applicable sections of MTP 2-2-502 (automotive) and MTP 9-2-500 (crane).
b. Record measurements of the steering mechanism.

6.3.2.4 Maintenance Evaluation

a. For the automotive portion of the test item, data shall be collected and recorded as prescribed in applicable sections of MTP 2-2-503.
b. Record the following for the crane portion of the test item:

1) Scheduled maintenance
2) The adequacy of spare parts and tools
3) Equipment deficiencies, causes, and suggested or corrective action taken
4) The adequacy and accuracy of the technical and maintenance instructions provided by the manufacturer

6.3.2.5 Safety Hazards Evaluation

Record the following:

a. Conditions or procedures that might present a safety hazard, the cause of the hazard
b. Steps taken to alleviate the safety hazard(s)
c. Safety release recommendations

6.3.2.6 Human Factors Evaluation

Data shall be collected and recorded as prescribed in applicable sections of MTP 10-2-505.

6.3.2.7 Value Analysis

Record the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

a. Nonfunctional features
b. Costly features
c. Nice-to-have features
d. Comments of the test personnel and test director

6.4 DATA REDUCTION AND PRESENTATION

Data obtained during the conduct of the test will be summarized using charts and graphs as appropriate. The test data will be evaluated by determining the extent to which it meets the requirements of the technical characteristics and military specifications for the crane truck.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:54Z
Check the source to verify that this is the current version before use.